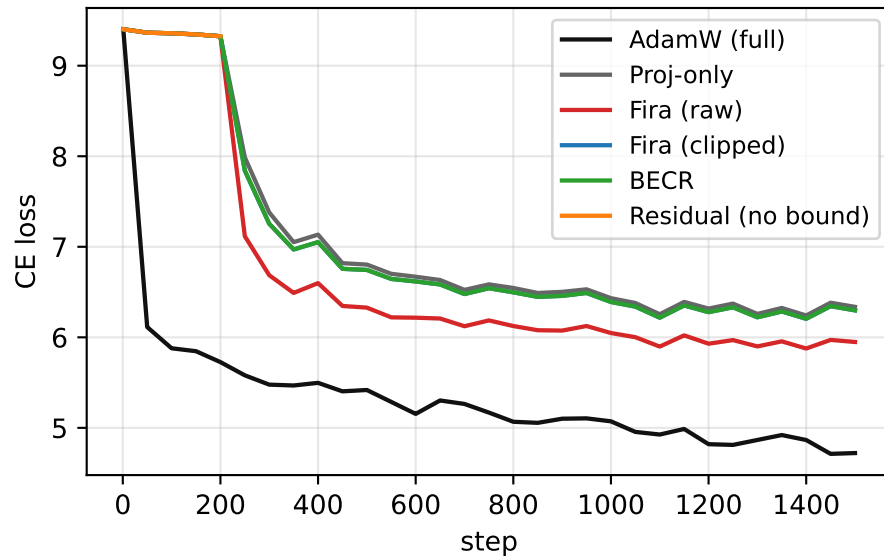
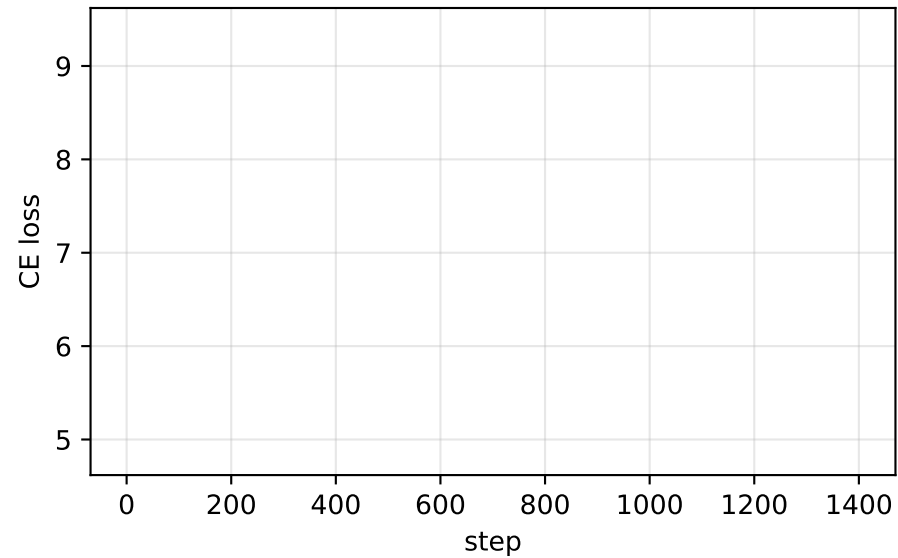


WikiText-2 (raw) TinyGPT (mechanism diagnostic)

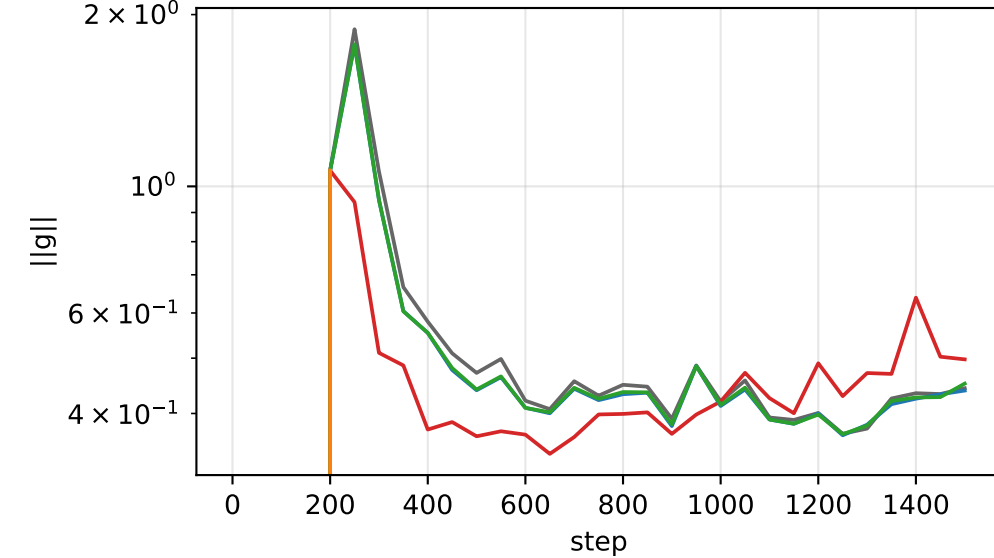
(a) Train loss (seed=0)



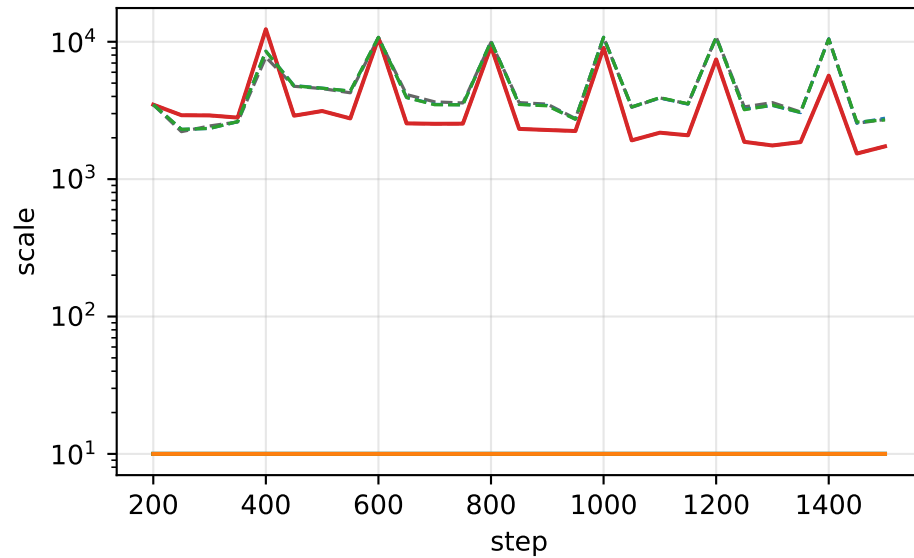
(b) Valid loss (seed=0)



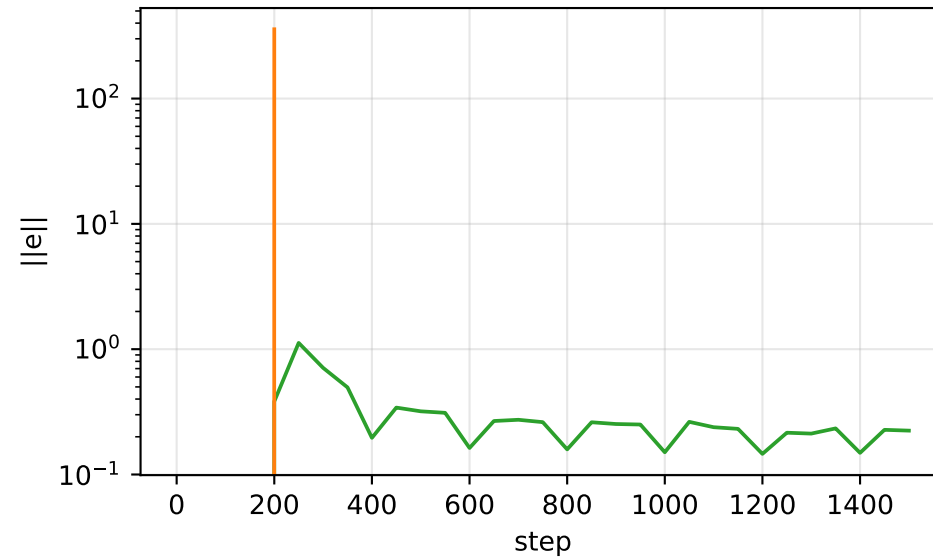
(c) $||\text{grad}||$ (logged, diagnostic)



(d) Scale p95: s_t (solid) vs ϕ_{raw} (dashed)



(e) Residual norm $||e||$ (BECR variants)



(f) Final valid perplexity (mean $\pm$ std over seeds)

